6 **(a)** *either* $P \propto V^2$ *or* $P = V^2/R$ ………………………………………………………… C1

reduction = $(230^2 - 220^2)/230^2$

= 8.5 % ……………………………………………………………………….. A1 [2]

(b) **(i)** zero …………………………………………………………………………… A1 [1]

(ii) 0.3(0) A ……………………………………………………………………….. A1 [1]

(c) **(i)** correct plots to within ± 1 mm ……………………………………………… B1 [1]

(ii) <u>reasonable line/curve</u> through points giving current as 0.12 A

allow ± 0.005A) …………………………………………………………… B1 [1]

(iii) $V = IR$ …………………………………………………………………… C1

$V = 0.12 \times 5.0$

$= 0.6(0)$ V ……………………………………………………………………... A1 [2]

(d) circuit acts as a potential divider/current divides/current in AC not the same as current in BC ………………………………………………………………………… B1

resistance between A and C not equal to resistance between C and B ……. B1

or current in wire AC × R is not equal to current in wire BC × R B1 [2]

any 2 statements

7 (a) (i) *either* helium nucleus